

OpenSR-SRGAN: A Flexible Super-Resolution Framework for Multispectral Earth Observation Data

Simon Donike¹, Cesar Aybar¹, Julio Contreras¹, and Luis Gómez-Chova¹

¹ Image and Signal Processing Group, University of Valencia, Spain ✉ Corresponding author

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Summary

Satellite images often contain useful measurements beyond visible red, green and blue light, but at a spatial resolution that is too coarse for some applications. Super-resolution methods estimate a finer-resolution image from these coarse observations. OpenSR-SRGAN is a modular, configuration-driven framework for training and benchmarking such models on multispectral Earth-observation data. The software enables users to train and evaluate configurable generator–discriminator architectures on arbitrary sensor band setups using concise YAML configurations, without modifying model code. All components, such as generators, discriminators, loss functions, training schedules, normalizations, and stability heuristics, are exposed through a common YAML interface. OpenSR-SRGAN makes it straightforward to reproduce experiments, compare architectures, and deploy super-resolution pipelines across diverse remote-sensing datasets.

OpenSR-SRGAN supports complete end-to-end workflows with minimal setup: selecting architectures, scaling factors, band combinations, and training strategies entirely from configuration files. Although designed for remote-sensing super-resolution, its tensor-based model components can be adapted to other paired imaging data when suitable preprocessing and validation are provided.

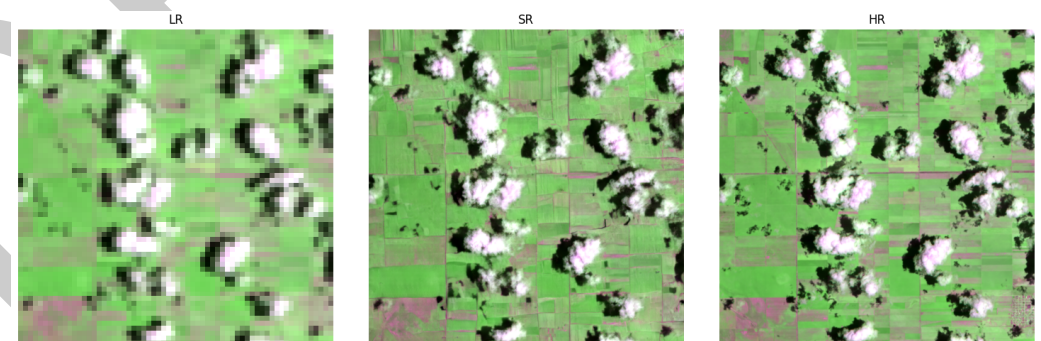


Figure 1: Sentinel-2 SWIR-band composite for 8x SR task showing the SR performance of OpenSR-SRGAN.

Introduction

Optical satellite imagery supports many geospatial applications, including agriculture (Weiss et al., 2020), land-cover mapping (Rogan & Chen, 2004), ecosystem assessment (Li et al., 2014) and disaster monitoring (Ahmad, 2024). Sensors such as Sentinel-2 provide rich multispectral

27 data but at moderate spatial resolution, motivating single-image super-resolution (SISR) to
28 recover finer detail from coarse observations.

29 Deep learning has driven major progress in SISR, with convolutional models improving fidelity
30 and perceptual quality (Dong et al., 2016; Kim et al., 2016). GANs (Goodfellow et al., 2014)
31 introduced adversarial learning for synthesizing realistic high-frequency detail and are widely
32 used in remote sensing (Donike, Aybar, Contreras, et al., 2025; Su et al., 2024). SRGAN (Ledig
33 & others, 2017) extended these ideas to super-resolution and has been applied to multispectral
34 data (Jia et al., 2022; Meng et al., 2024; Wang et al., 2023; Wolters et al., 2023).

35 Although diffusion and transformer-based models increasingly define the state of the art
36 (Donike, Aybar, Gómez-Chova, et al., 2025; Liu et al., 2022; Xiao et al., 2024), GAN-based
37 SR methods remain relevant for efficient and deterministic enhancement workflows (Panangian
38 & Bittner, 2025; Wolters et al., 2023).

39 Statement of Need

40 GANs remain difficult to train (Bau et al., 2019; Bowles et al., 2018; Salimans et al., 2016),
41 and these challenges are amplified in remote sensing, where models must handle multispectral
42 inputs, high dynamic-range reflectance values, heterogeneous sensor characteristics, and
43 limited availability of aligned high-resolution reference data. Many research implementations
44 reproduce a single architecture or assume RGB imagery, offering limited flexibility for modifying
45 band configurations, normalization schemes, loss compositions, or training strategies. As a
46 result, researchers who need to adjust models for different sensors (e.g., Sentinel-2, SPOT,
47 Pleiades, or PlanetScope) must often re-engineer core components, modify low-level code, and
48 manually implement stabilization heuristics such as warmup, ramping, or EMA tracking, as
49 well as logging mechanisms. This makes reproducing published experiments or conducting
50 systematic comparisons across architectures labor-intensive, brittle, and inconsistent across
51 studies. These challenges call for a flexible, extensible and configuration-first framework
52 that reduces implementation overhead while enabling systematic experimentation across
53 architectures, loss designs and multispectral sensor configurations. The primary audience is
54 researchers and practitioners developing, comparing, or deploying multispectral super-resolution
55 models for Earth-observation data.

56 State of the Field

57 General-purpose restoration frameworks such as BasicSR¹ and MMagic² provide configurable
58 implementations of SRGAN, ESRGAN, and other super-resolution models. TorchGeo provides
59 complementary datasets, samplers, and multispectral utilities for geospatial machine learning
60 (Stewart et al., 2022). The BasicSR package is broad, but does not provide the complete
61 combination of multispectral GAN configurations, reflectance-aware normalization, spectral
62 losses, stabilization schedules, Sentinel-2 presets, and evaluation and deployment interfaces
63 necessary for remote-sensing specific workflows.

64 OpenSR-SRGAN was developed as a separate focused package because its contribution is this
65 shared Earth-observation experimentation protocol rather than a new generator architecture.
66 Implementing the same functionality as isolated additions to a general restoration framework
67 would still leave researchers responsible for assembling and maintaining the sensor-specific
68 training, evaluation, and large-scene inference workflow. OpenSR-SRGAN instead integrates
69 these choices in one compact interface while reusing established libraries and the wider OpenSR
70 ecosystem.

¹<https://github.com/XPiPixelGroup/BasicSR>

²<https://github.com/open-mmlab/mmagic>

Software Design

OpenSR-SRGAN provides a unified, modular and configuration-driven framework for training and evaluating GAN-based super-resolution models for multispectral remote sensing data. All components of an experiment, including architectures, losses, optimizers, data pipelines and training behavior, are defined through concise YAML files, which makes runs repeatable and auditable without modifying source code. The framework supports arbitrary band configurations and adapts naturally to different sensors.

The configuration-first design was chosen to make comparisons auditable: generators, discriminators, losses, and training schedules are instantiated through shared factories. Flat YAML configurations remain easy to archive, while Hydra-based composition supports reusable presets and command-line overrides. This design favors a stable and inspectable experiment schema.

The main features include:

- **Modular GAN framework:** Interchangeable generator and discriminator backbones with configurable depth, width and scale factors.
- **Configuration-first workflow:** Repeatable training and evaluation using concise YAML definitions.
- **Training stabilization options:** Warmup, adversarial ramping, label smoothing, spectral normalization, adaptive learning-rate scheduling and optional EMA tracking.
- **Multispectral compatibility:** Native support for arbitrary band combinations across sensors.
- **OpenSR ecosystem integration:** Standardized evaluation via `opensr-test` (Aybar, Montero, Donike, et al., 2024) and scalable inference utilities via `opensr-utils` (Donike & contributors, 2025).
- **HPC-ready inference:** Inference scripts for HPC-clusterse enabling large-scale SR product generation.

A typical workflow in OpenSR-SRGAN involves selecting an architecture, defining the desired scale factor, specifying multispectral band ordering and choosing appropriate loss terms, all through simple YAML file modifications. The framework then handles model construction, dataset loading, training procedures and evaluation, which allows researchers to focus on experiment design rather than implementation details. This approach is particularly useful when comparing multiple configurations or running ablation studies, since each experiment is defined through its configuration file. Internally, the framework supports flexible loss composition, allowing pixel and spectral fidelity to be balanced against perceptual sharpness and adversarial realism. PyTorch Lightning's manual-optimization path was chosen to make generator and discriminator update order, pretraining, adversarial ramping, and EMA updates explicit, at the cost of not using Lightning's automatic optimizer handling. Users may also enable Wasserstein GAN training (Arjovsky et al., 2017) with an optional R1 gradient penalty (Mescheder et al., 2018). These components together form a practical and extensible foundation for SRGAN experimentation while keeping the central training trade-offs visible.

Research Impact Statement

Version 0.5.1 is distributed through the Python Package Index and archived on Zenodo. The repository provides user and API documentation, a runnable Colab example, contribution guidelines, and automated tests exercised by continuous integration on Python 3.12, 3.13, and 3.14. Resolved experiment configurations, example data, pretrained weights, and a Slurm-based large-area inference workflow provide concrete reproducibility and community-readiness

120 evidence beyond the software source alone.

121 OpenSR-SRGAN is the adversarial-model component of the European Space Agency OpenSR
122 project. It includes a loader for the published SEN2NAIP dataset (Aybar, Montero, Contreras,
123 et al., 2024), produces models that can be evaluated with opensr-test (Aybar, Montero,
124 Donike, et al., 2024), and integrates opensr-utils (Donike & contributors, 2025) for tiled
125 inference on large Sentinel-2 products. Public pretrained configurations and checkpoints support
126 ongoing experiments on four-band RGB–NIR and six-band short-wave-infrared super-resolution.

127 Limitations

128 Super-resolution methods enhance apparent detail but cannot replace imagery collected at
129 native high resolution. OpenSR-SRGAN focuses on flexibility and reproducibility rather than
130 state-of-the-art performance, and results depend on proper preprocessing and accurate LR–HR
131 alignment. GAN training and SR in general remain sensitive to dataset size and diversity, and
132 may produce instability or spectral artifacts, particularly when reference data are geographically
133 skewed, biased by land cover, or limited in size and scope.

134 Licensing and Availability

135 The source code is made available through the [ESAOpenSR/OpenSR-SRGAN](#) Github repository.
136 Full documentation, API references, quickstart guides and tips and tricks can be found at
137 [srgan.opensr.eu](#). A reproducible notebook is permanently hosted on [Google Colab](#). In the
138 spirit of open science and collaboration, we encourage feature requests and updates, bug fixes
139 and reports, as well as general questions and concerns via direct interaction with the repository.
140 OpenSR-SRGAN is licensed under the Apache-2.0 license.

141 AI Usage Disclosure

142 OpenAI Codex (versions 5.1-5.5) was used to assist with and generate parts of the
143 documentation, automated tests and continuous-integration workflows, and model training and
144 implementation code. The authors reviewed all AI-assisted changes and retain responsibility
145 for the final code and functionality. Software changes are verified through targeted tests, the
146 continuous-integration suite, and frequent manual usage. Experimental data, results, imagery,
147 and pretrained model weights are fully reproducible. The manuscript has been edited for
148 clarity and readability only, using Writeful embedded in Overleaf.

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